

Homework 1 - Questions and Study Solutions

EEE 554 - Probability and Random Processes | Fall 2026

Prepared as a learning reference. Review each derivation before writing your own submission.

Notation note: $P(A \text{ intersect } B)$ means the probability that both A and B occur. The solutions use the probability rules and Bayes' theorem covered in Module 1, along with the continuous-uniform model introduced in Module 2.

1. Uniform reboot time (5 points)

Question

A computer is programmed to reboot automatically for updates between 4:10 am and 4:55 am. The computer is equally likely to reboot anytime within this time interval.

- (a) What is the probability that the computer will reboot at exactly 4:16 am?
- (b) What is the probability that the computer will reboot between 4:15 am and 4:30 am?

Solution

Let T be the reboot time in minutes after 4:10. Then T is continuous and uniform on $[0, 45]$.

- (a) $P(T = 6) = 0$. A continuous random variable assigns probability zero to any one exact time.
- (b) The interval from 4:15 to 4:30 has length 15 minutes. Therefore $P(5 \leq T \leq 20) = 15/45 = 1/3$.

2. Events in a two-dimensional sample space (15 points)

Question

The sample space $S = \{(x, y): 0 \leq x \leq 3, 0 \leq y \leq 3\}$ includes values of x, y in the two-dimensional Cartesian space. Events are $A = \{(x, y): y \leq 3 - x\}$ and $B = \{(x, y): y \geq x\}$.

- (a) Draw the regions corresponding to events A and B in the (x, y) -plane.
- (b) Evaluate $P(A)$, $P(B)$, and $P(A | B)$.
- (c) Are A and B independent? Justify your answer.

Solution

The sample space is the 3 by 3 square, so its area is 9. A is the triangular region below $y = 3 - x$; B is the triangular region above $y = x$.

- (a) Draw the square with corners $(0,0)$, $(3,0)$, $(3,3)$, and $(0,3)$. Draw $y = 3 - x$ from $(0,3)$ to $(3,0)$, shading below it for A. Draw $y = x$ from $(0,0)$ to $(3,3)$, shading above it for B.
- (b) Each triangle has area $(1/2)(3)(3) = 9/2$, so $P(A) = P(B) = (9/2)/9 = 1/2$. The intersection $A \text{ intersect } B$ satisfies $x \leq y \leq 3 - x$, which requires $0 \leq x \leq 3/2$. Its area is integral from 0 to $3/2$ of $[(3 - x) - x] dx = 9/4$. Hence $P(A \text{ intersect } B) = (9/4)/9 = 1/4$ and $P(A | B) = (1/4)/(1/2) = 1/2$.
- (c) Yes. $P(A \text{ intersect } B) = 1/4 = (1/2)(1/2) = P(A)P(B)$. Equivalently, $P(A | B) = P(A) = 1/2$.

3. Discrete tardiness distribution (10 points)

Question

A habitually tardy person arrives at the theater late by s_i minutes, where $s_i = i$ for $i = 1, 2, 3, \dots$. If $P[s_i] = (1/2)^i$, what is the probability that they will be more than one minute late?

Solution

$P(\text{more than one minute late}) = P(s_2) + P(s_3) + \dots = \text{sum from } i=2 \text{ to infinity of } (1/2)^i$.

This geometric series has first term $1/4$ and ratio $1/2$, so its sum is $(1/4)/(1 - 1/2) = 1/2$. Equivalently, it is $1 - P(s_1) = 1 - 1/2 = 1/2$.

4. Drawing with replacement (10 points)

Question

An urn contains 3 red balls and 2 black balls. Two balls are chosen in succession. The first ball is returned to the urn before the second ball is chosen. Each ball is chosen at random.

- (a) What is the probability of choosing first a red ball and then a black ball?
- (b) More generally, if the urn has k red balls and $N - k$ black balls, what is the probability of a red ball followed by a black ball when two balls are chosen at random with replacement?

Solution

Because the first ball is replaced, the two draws are independent.

(a) $P(\text{red then black}) = P(\text{red})P(\text{black}) = (3/5)(2/5) = 6/25$.

(b) $P(\text{red then black}) = (k/N)((N - k)/N) = k(N - k)/N^2$.

5. Shared birthday (10 points)

Question

A probability class has N students enrolled. What is the probability that at least two of the students will have the same birthday?

Solution

Use the complement: subtract the probability that all N birthdays are distinct from 1. Assuming 365 equally likely birthdays and ignoring leap day, $P(\text{all distinct}) = (365/365)(364/365)\dots((365 - N + 1)/365)$.

Therefore $P(\text{at least one shared birthday}) = 1 - \text{product from } j=0 \text{ to } N-1 \text{ of } (365 - j)/365$, for $N \leq 365$. For $N > 365$, the probability is 1.

6. Independence with a fair die (10 points)

Question

Given a fair die, consider A = getting 2 or 3 and B = getting an even number. Are A and B independent? (Hint: compare $P[A]$ and $P[A | B]$.)

Solution

$A = \{2, 3\}$, so $P(A) = 2/6 = 1/3$. $B = \{2, 4, 6\}$. Given B , only 2 of the three even outcomes is in A , so $P(A | B) = 1/3$.

Since $P(A | B) = P(A)$, A and B are independent. Also, $P(A \cap B) = P(\{2\}) = 1/6 = (1/3)(1/2)$.

7. Conditional events with a fair die (10 points)

Question

Toss a fair die and consider A = getting an even number; B = getting a number greater than 2; and C = getting a number less than 5. Find $P[C]$, $P[B | C]$, $P[A | B, C]$, and $P[ABC]$.

Solution

$C = \{1, 2, 3, 4\}$, so $P(C) = 4/6 = 2/3$.

$B \cap C = \{3, 4\}$; therefore $P(B | C) = 2/4 = 1/2$.

$B \cap C = \{3, 4\}$, and its even outcome is $\{4\}$; therefore $P(A | B, C) = 1/2$.

$A \cap B \cap C = \{4\}$, so $P(ABC) = 1/6$.

8. Cancer-test Bayes calculation (10 points)

Question

Suppose 0.001% of the general population has a certain cancer. A blood test is positive with probability 0.99 if a patient has the cancer. It produces a false positive with probability 0.2 if the patient does not have cancer. If the test comes back positive, what is the probability that the person has cancer?

Solution

Let C mean cancer and $+$ mean a positive test. The prevalence is $P(C) = 0.001\% = 0.00001$. Given values: $P(+ | C) = 0.99$ and $P(+ | C^c) = 0.2$.

By Bayes' theorem, $P(C | +) = [P(+ | C)P(C)] / [P(+ | C)P(C) + P(+ | C^c)P(C^c)]$.

Thus $P(C | +) = (0.99)(0.00001) / [(0.99)(0.00001) + (0.2)(0.99999)] = 0.0000099 / 0.2000079 = \text{approximately } 0.00004950$.

The probability is approximately 0.00495%, or about 1 in 20,202. The low prevalence and relatively high false-positive rate dominate the result.

9. Repeated fax dialing (10 points)

Question

A fax machine dials a phone number that is typically busy 80% of the time. It dials every 5 minutes until the line is clear and the fax can be transmitted. What is the probability that it will have to dial the number 9 times?

Solution

A ninth dial means that the first eight dials find a busy line and the ninth finds a clear line. With $P(\text{busy}) = 0.8$ and $P(\text{clear}) = 0.2$, this is a geometric distribution event.

$P(\text{ninth dial}) = (0.8)^8(0.2) = 0.033554432$, approximately 3.36%.

10. Forming the word committee (10 points)

Question

A person chooses 9 letters at random from the English alphabet with replacement. What is the probability that they will be able to make the word committee?

Solution

The word committee has 9 letters with repeated letters m, t, and e each appearing twice. There are $9!/(2! 2! 2!) = 45,360$ distinct sequences that can be rearranged into committee.

Each ordered sequence of nine independent letters has probability $(1/26)^9$. Therefore $P = [9!/(2! 2! 2!)](1/26)^9 = 45,360/26^9$, approximately 8.35×10^{-9} .